



Hybrid integrated thin-film lithium niobate–silicon nitride electro-optical phased array incorporating silicon nitride grating antenna for two-dimensional beam steering

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Abstract: This study proposes a solid-state two-dimensional beam-steering device based on an electro-optical phased array (EOPA) in thin-film lithium niobate (TFLN) and silicon nitride (SiN) hybrid platforms, thereby eliminating the requirement for the direct etching of TFLN. Electro-optic (EO) phase modulator array comprises cascaded multimode interference couplers with an SiN strip-loaded TFLN configuration, which is designed and fabricated via i-line photolithography. Each EO modulator element with an interaction region length of 1.56 cm consumed a minimum power of $3.2 \text{ pJ}/\pi$ under a half-wave voltage of 3.64 V and had an estimated modulation speed of 1.2 GHz. Subsequently, an SiN dispersive antenna with a waveguide grating was tethered to the modulator array to form an EOPA, facilitating the out-of-plane radiation of highly defined near-infrared beams. A prepared EOPA utilized EO phase control and wavelength tuning near 1550 nm to achieve a field-of-view of $22^\circ \times 5^\circ$ in the horizontal and vertical directions. The proposed hybrid integrated platform can potentially facilitate low-power and high-speed beam steering.

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1. Introduction

Optical phased arrays (OPAs) are a prominent solid-state alternative to the conventional bulky motor- and microelectromechanical systems (MEMS)-based beam-steering devices owing to their compactness, low cost, high reliability, free inertia, and fast speed. They are applicable in light detection and ranging (LiDAR), free-space optical communication, and imaging [1–5]. OPA devices based on silicon and silicon nitride (SiN) photonic platforms have emerged in view of their complementary metal-oxide-semiconductor (CMOS) compatibility, providing a wide field-of-view (FOV) that is enhanced by aliasing-free scanning, small beamwidth, and large-scale arrays with thousands of elements [6–10]. However, the reported silicon and SiN OPAs depend on the thermo-optic effect to achieve phase tuning, resulting in low beam-steering speed and high power consumption. To overcome these limitations, electro-optic (EO) integrated circuit schemes based on doped silicon, indium phosphide, and EO polymers have been developed [11–13]. However, these materials are susceptible to doping-level-sensitive absorption losses, temporal stability, and CMOS incompatibility [14–16]. Recent advancements in high-quality thin-film lithium niobate (TFLN) have paved the way for embedding compact waveguides to yield a higher index contrast and stronger optical confinement compared to bulk lithium niobate [17,18]. In contrast to conventional modulators, TFLN is an attractive EO material because it exhibits high-speed modulation, low power consumption, broad transparency window, and low propagation loss while maintaining a high degree of integration [19,20]. TFLN modulators operate with drivable voltages in the CMOS circuitry and ultra-high bandwidths [21,22]. Therefore, the TFLN platform was utilized for the implementation of high-performance OPA chips [23,24]. However, the patterning of TFLN unavoidably requires a sophisticated and scalable etching technique

imposing a strict process control, while lithium element contained in TFLN may be regarded as a contaminant in CMOS processing [14,18]. An SiN strip-loaded TFLN (TFLN–SiN) platform circumvents the direct etching of TFLN, offering numerous advantages such as CMOS-compatible SiN etching for patterning and remarkably low optical propagation loss [25,26]. The platform, which enables GHz-level fast beam steering and low power consumption for beamforming, can provide an advanced modulation modality to OPAs. However, the strip-loaded waveguide structure may adversely affect the beam-steering performance. The TFLN–SiN waveguide has a laterally weak confined mode, which mostly engenders crosstalk between adjacent waveguides, generating a disturbance factor that reduces the gap between waveguides. To improve the radiation performance, the etch depth and duty cycle of the grating, which is an SiN strip, should be tailored to modify the leakage factor [27]. Therefore, the emitting components require a more suitable platform instead of TFLN–SiN. SiN has emerged as a prominent candidate, offering salient advantages such as low propagation loss, extended transparent band, low nonlinearity, and high tolerance to fabrication errors due to low index contrast. Our previous study reported that an SiN waveguide provides strong mode confinement that allows a minimum channel spacing of 3 μm . The steering performance can be improved using a strong grating vector, bidirectional grating antenna, optical lenses, and metalens [5,10,28,29]. Hence, an SiN antenna can be exploited to impart beam radiation functionality in free space.

This study proposes a hybrid integrated TFLN–SiN electro-optical phased array (EOPA) incorporating an SiN grating antenna. The EO phase modulator, which is constructed using an TFLN–SiN platform, plays the role of coping with the slow modulation speed and high power consumption of thermo-optic phase modulators that comprise a majority of the reported OPAs. The EO modulator array is configured using a power splitter and phase modulators. The power splitter is realized by serially concatenating multimode interference (MMI) couplers. The constituting phase modulator is rigorously investigated in terms of the modulation bandwidth, driving voltage, power consumption, and fabrication tolerance. In an attempt to impart the capability of wavelength-tuned beam radiation, a partially etched SiN grating waveguide is deployed as an antenna and butt-coupled to the TFLN–SiN modulator array. For OPA devices with and without the SiN grating antenna, the two-dimensional (2D) steering is executed by means of the phase tuning and wavelength tuning in the horizontal and vertical directions, respectively. To the best of our knowledge, our study is the first to demonstrate an EOPA using the proposed hybrid TFLN–SiN platform.

2. Design and fabrication of the proposed EOPA

Figure 1 shows the proposed EOPA comprising a TFLN–SiN EO modulator array and SiN grating antenna. The input light was distributed into 16 channels using a power splitter, and the phase for each channel was independently controlled by applying a voltage to the TFLN–SiN phase modulators via the EO effect. After navigating the transition bend, the phase-modulated light was launched out of the plane from an SiN grating antenna to execute 2D beam steering along the horizontal (ψ) and vertical (θ) directions. As shown in Fig. 2(a), the TFLN–SiN platform was constructed by loading an SiN strip atop an x-cut TFLN with a bottom oxide layer on a silicon substrate. The TFLN had a thickness H_{TLN} of 400 nm, and the lower oxide cladding height H_{SiO_2} was 2 μm . To support a fundamental transverse electric (TE) mode, the strip had a width W_{SiN} of 1.5 μm and height H_{SiN} of 250 nm. Subsequently, the SiN strip was encapsulated by the top oxide cladding with a height H_{Clad} of 200 nm. Figure 2(b) shows the TE mode profile of the waveguide that was calculated at 1550 nm through a finite difference eigenmode (FDE) solver in MODE Solutions (Ansys Lumerical), exhibiting a laterally loosely confined mode owing to the strip waveguide structure. When the spacing between waveguide channels is excessively reduced to increase the FOV, the relatively loosely confined mode is prone to engendering undesired crosstalk. The proposed EOPA employed the 1×2 MMI based 3-dB coupler to configure the

power splitter for even optical power distribution across multiple channels, as shown in Fig. 2(c), instead of using a Y-branch and directional coupler, which were sensitive to fabrication errors [30]. Considering the limited minimum feature size and spacing of photolithography, the MMI coupler was designed to be preferably large in size to minimize fabrication errors. The MMI width W_{MMI} was $14\ \mu\text{m}$, and the output ports were separated by a distance S . The footprint of the MMI coupler was determined by sweeping the MMI length L_{MMI} to achieve low insertion loss using a three-dimensional (3D) eigenmode expansion solver in MODE Solutions (Ansys Lumerical) for optimization. This yielded a throughput loss less than 0.1 dB at the wavelength of 1550 nm under L_{MMI} of $145\ \mu\text{m}$, tantamount to the length based on the self-imaging method [31]. In the operating spectral band ranging from 1530 to 1600 nm, the MMI coupler approximately exhibited 0.1-dB variation in excess loss. Considering the minimum taper length of input/output taper for maximum transmission, the values of L_{taper} and W_{taper} were set to be $100\ \mu\text{m}$ and $4\ \mu\text{m}$, respectively. Figure 2(d) shows the calculated electric field profile pertaining to the MMI coupler, where the final design parameters are $W_{\text{WG}} = 1.5\ \mu\text{m}$, $W_{\text{MMI}} = 14\ \mu\text{m}$, $L_{\text{MMI}} = 145\ \mu\text{m}$, $W_{\text{taper}} = 4\ \mu\text{m}$, $L_{\text{taper}} = 100\ \mu\text{m}$, and $S = 7\ \mu\text{m}$. The MMI power splitter evenly distributed the input light into 16 channels with a spacing of $18\ \mu\text{m}$ in four stages.

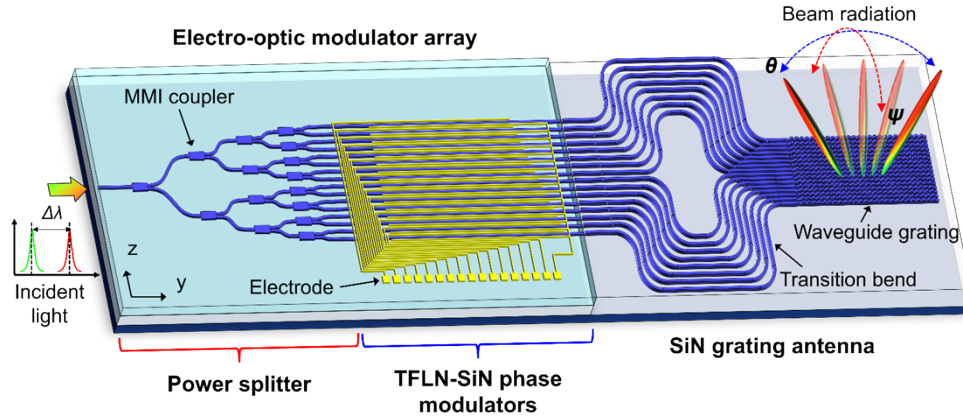


Fig. 1. Configuration of the proposed hybrid integrated EOPA, which incorporates an EO modulator array on a TFLN–SiN platform that is in conjunction with a dispersive SiN grating antenna.

An EO phase modulator is an essential component for implementing low-power and high-speed beam steering. For the TFLN–SiN phase modulator, a gold electrode was placed on the top cladding of the TFLN–SiN waveguide to vary the refractive index via the EO effect, as shown in Fig. 3(a). The thickness H_{Au} and width W_{Au} of the electrodes were $300\ \text{nm}$ and $4\ \mu\text{m}$, respectively. The electrode gap g was $7\ \mu\text{m}$ to avoid the ohmic loss caused by metals owing to photolithography alignment error. The cladding was deposited beneath the electrode to prevent the ohmic loss resulting from the electrode atop the waveguide and possible contamination caused by external contact. The propagation direction of the strip waveguides is perpendicular to the z -axis, while the modulating electric field pertaining to the electrodes aligns along the principal z -axis of TFLN. Figure 3(b) shows the optical TE mode field and calculated electric field distribution when a constant voltage was applied to the electrodes. The strategic placement of the integrated electrodes is favorable for obtaining a maximum EO coefficient of r_{33} for the TE mode. The mode confinement factor in the TFLN of the proposed structure was $\sim 68\%$. The narrower the gap between electrodes placed on either side of the strip waveguide, the smaller the half-wave voltage V_{π} of the phase modulator. The half-wave voltage and length product derived from the COMSOL-based simulations yielded a value of $5.92\ \text{V}\cdot\text{cm}$ with approximately zero propagation

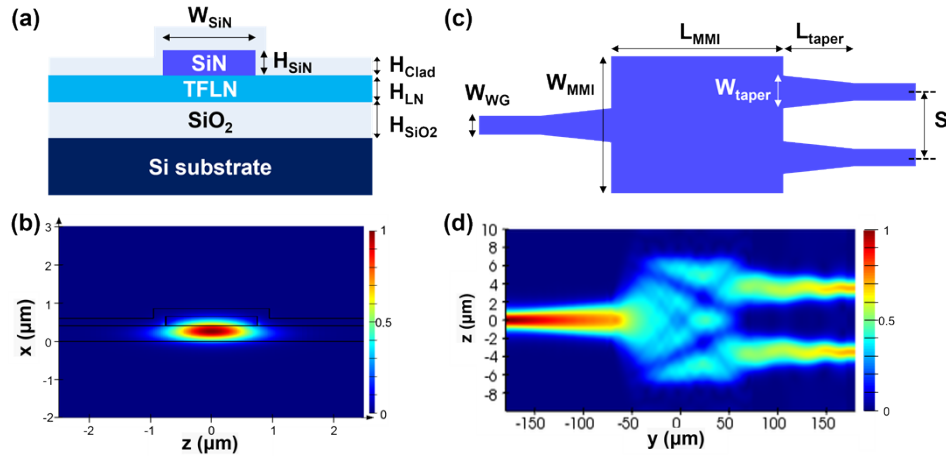


Fig. 2. (a) Cross-sectional view of the TFLN–SiN waveguide. (b) Simulated optical TE mode profile of the TFLN–SiN waveguide. (c) Schematic of the TFLN–SiN 1×2 MMI coupler. (d) Simulated electric field distribution of the MMI coupler along the propagation direction.

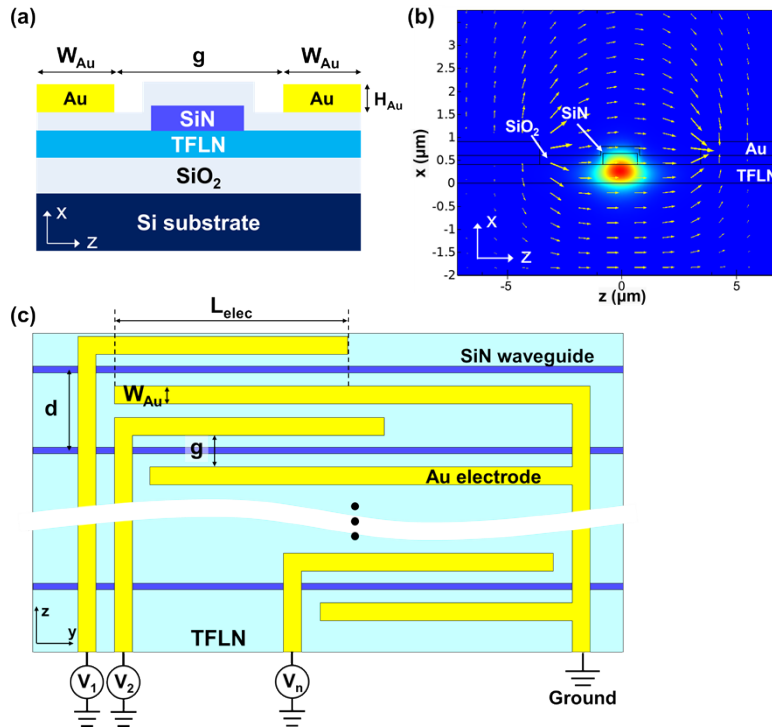


Fig. 3. (a) Cross-sectional view of the TFLN–SiN phase modulator. (b) Calculated optical mode and electric field distribution when 1 V is applied to the modulation electrodes. (c) Top view of the TFLN–SiN phase modulator array.

loss. For horizontal beam steering, the modulators were arrayed at a spacing d of $18 \mu\text{m}$ and EO interaction length L_{elec} of 1.56 cm , as depicted in Fig. 3(c). The RC-limited modulation bandwidth was calculated up to 1.2 GHz for a 1.56 cm -long electrode with an estimated capacitance of 0.63

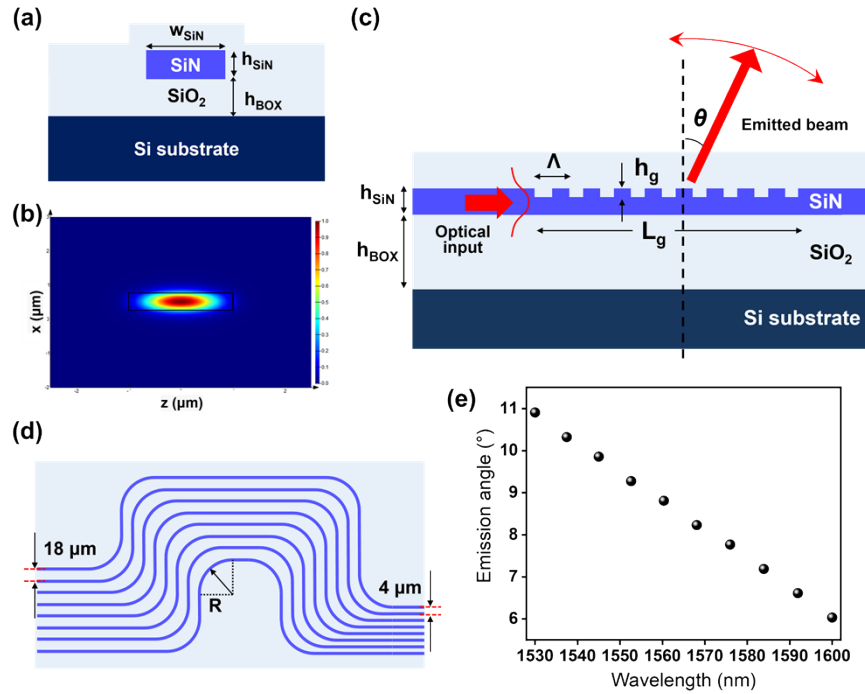


Fig. 4. (a) Cross-sectional view of the SiN waveguide structure. (b) Optical TE mode profile of the SiN waveguide. (c) Configuration of the SiN grating antenna and its operation principle. (d) Schematic of the transition bend to reduce the channel spacing while maintaining an equal length. (e) Simulated beam emission angle versus the wavelength.

pF. The operation of the TFLN–SiN phase modulator, serving as a capacitor, was based on the EO effect. The electrical energy required for inducing a phase shift of π can be estimated based on $E = CV_p^2/4$, where C is the modulator capacitance and V_p is the peak half-wave modulation voltage, indicating a power consumption of $2.3 \text{ pJ}/\pi$ per phase modulator.

The SiN waveguide exhibited stronger optical mode confinement compared to the TFLN–SiN waveguide, thereby realizing denser waveguide arrays to enhance the FOV of the TFLN–SiN EOPA. As shown in Fig. 4(a), the SiN waveguides were constructed as an SiN core surrounded by oxide, which has a height h_{SiN} of 500 nm and width w_{SiN} of 2 μm , on top of a buried oxide layer with a thickness h_{BOX} of 4 μm deposited on a silicon substrate which is covered with a 3 μm -thick oxide cladding. Figure 4(b) portrays the calculated optical TE mode of the SiN waveguide, exhibiting a laterally stronger mode confinement compared to the TFLN–SiN waveguide. The radiated beam quality and steering performance could be enhanced by changing the waveguide platform using a higher mode confinement. To achieve beam steering along the vertical direction based on wavelength tuning, the antenna incorporated a partially etched surface grating with a length L_g of 500 μm , period Λ of 1 μm , fill factor of 0.5, and grating depth h_g of 100 nm, as shown in Fig. 4(c). The 16-channel SiN waveguide array was arranged with a uniform spacing of 18 μm to accommodate the phase-modulated modes. Figure 4(d) reveals that the spacing is gradually reduced from 18 μm to 4 μm to enlarge the FOV with the aid of the transition bend. The SiN grating antenna, featuring a 2- μm width, has been designed to have a channel spacing of 4 μm , from the perspective of effectively suppressing channel crosstalk over a length of 500 μm . Introducing an equivalent optical path length across the channels helps prohibit unnecessary phase difference $\Delta\phi$ between channels owing to wavelength tuning [10]. The radius R of the 90° bent waveguide was fixed at 75 μm to suppress the optical bending loss. Finally,

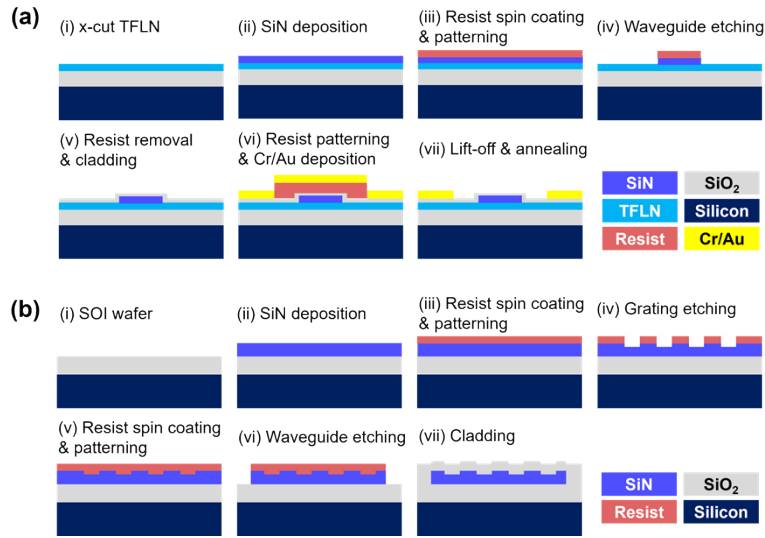


Fig. 5. Schematic of the fabrication flow of (a) the TFLN-SiN waveguides and phase modulators and (b) the SiN waveguide and grating antenna.

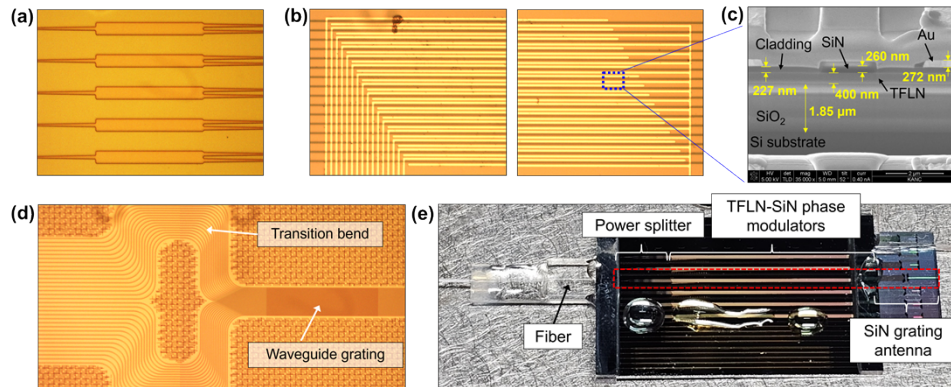


Fig. 6. Optical microscope and SEM images of the fabricated devices. (a) Optical microscope image of the 1 × 2 MMI coupler. (b) Optical microscope image and (c) SEM image of the phase modulator. (d) Optical microscope image of the SiN grating antenna. (e) Optical image of the proposed hybrid integrated TFLN-SiN EOPA incorporating the SiN grating antenna.

the grating structure diffracted the modulated light from the waveguide into free space. The emission angle θ is determined by $\sin \theta = n_{\text{eff}} - \lambda/\Lambda$, where n_{eff} is the effective refractive index of the SiN waveguide and λ is the wavelength. As shown in Fig. 4(e), a 3D finite-difference time-domain based tool (Ansys Lumerical) was used to numerically calculate the emission angle at various wavelengths in the range of 1530 to 1600 nm. The steering range in response to a given wavelength tuning range of 70 nm was 5°, leading to a beam-steering efficiency of approximately 0.07°/nm. To check the coupling loss between the TFLN-SiN and SiN waveguides, the mode field diameter (MFD) and coupling efficiency were observed using the FDE solver. As can be found in Figs. 2(b) and 3(b), the MFDs of the TFLN-SiN and SiN waveguides were $2.15 \times 0.72 \mu\text{m}^2$ and $1.74 \times 0.75 \mu\text{m}^2$, respectively; the coupling efficiency between the two waveguides was about 96%.

The designed TFLN–SiN waveguides and phase modulators were fabricated according to a standard CMOS process, as elaborately delineated in Fig. 5(a). To create a strip-loaded structure, an SiN layer was deposited on TFLN via plasma enhanced chemical vapor deposition (PECVD). A contact aligner was used to define the waveguide and modulator on an SiN thin film via i-line photolithography. Inductively coupled plasma (ICP) etching was conducted to etch the SiN layer, and the patterned waveguides were covered with a 200 nm-high oxide cladding via PECVD. Subsequently, a 300 nm-thick Cr/Au layer was evaporated on the cladding via electron beam evaporation, and a lift-off process was performed to form the signal ground configuration. An annealing process was subsequently conducted at 350°C for 1.5 hours to abate the optical absorption loss by removing the residual hydrogen of the SiN film without inducing cracks in the TFLN. As a result, a propagation loss of 2.03 dB/cm has been obtained from the fabricated TFLN–SiN waveguide. Microscopic images of the fabricated EO phase modulator array are displayed in Figs. 6(a) and (b). Figure 6(c) shows a scanning electron microscopy (SEM) image of the phase modulator, which satisfactorily emulates the designed structure. The fabrication process of the SiN waveguides and grating antenna, discussed in previous studies [5,10], is schematically illustrated in Fig. 5(b). The SiN layer was deposited via low-pressure chemical vapor deposition, resulting in low propagation loss [32]. A waveguide channel and grating emitter were constructed with the help of a deep ultraviolet stepper and ICP etching. Subsequently, an oxide cladding was deposited as a protective layer via PECVD. The optical microscope image of the SiN-grating antenna is displayed in Fig. 6(d). The fabricated TFLN–SiN and SiN platform wafers were subjected to precise dicing and polishing. Finally, the two chips were bonded via UV epoxy, and the contact area between the chips was expanded by placing a glass lid to prevent misalignment while curing, as shown in Fig. 6(e).

3. Characterization of the fabricated EOPA

Before manufacturing the hybrid integrated chip, the characteristics of the TFLN–SiN EO modulator array were primarily assessed in terms of the MMI coupler efficiency, half-wave voltage and length product value of the phase modulator, and beam-steering performance. To characterize the MMI coupler, a tunable laser (Santec, WSL-110) was deployed to launch the light of TE mode into the device passing via high numerical aperture fiber (Thorlabs, UHNA7) with an MFD of 3 μm . The light output was delivered to a UHNA7 fiber, and an optical power meter (Thorlabs, PM400) was connected to detect the light. The MMI power splitter comprised four-stage cascaded MMI couplers that contain 2^n channels for each stage. The coupling efficiency of the MMI power splitter was measured for each stage and plotted in Fig. 7(a). The excess loss for a constituting MMI coupler was measured in the operating wavelength range of 1530 to 1600 nm, as shown in Fig. 7(b). For each constituting MMI coupler, the excess loss was measured to be approximately 0.18 dB at a wavelength of 1550 nm, leading to an average value of 0.23 dB over the spectral band. The near-field guided mode profile pertaining to the phase modulator array was measured by an SWIR camera (AVAL GLOBAL, ABA-001IR), as shown in Fig. 7(c). The 4 μm -wide gold wire is passing over the strip waveguide to connect to a contact pad above the cladding, engendering a negligibly small ohmic loss of 0.1 dB, as illustrated in Fig. 3(c).

To scrutinize its modulation performance, the proposed EO phase modulator was practically evaluated using a Mach–Zehnder interferometer, which incorporated a push–pull electrode configuration in conjunction with an MMI coupler serving as a power divider and combiner. Figure 8(a) shows the experimental setup. Before propagating through the modulator, a 1550 nm-wavelength light was manipulated using a polarization controller and amplified through an erbium-doped fiber amplifier (EDFA). An arbitrary waveform generator was employed to supply a saw-tooth voltage wave with a low frequency of 100 kHz, which was fed to the electrodes using a ground-signal-ground (GSG) probe (GGB Industries Inc., Model 40A). The optical output was converted to an electrical signal using a photoreceiver (Newport, model 1592) tethered to

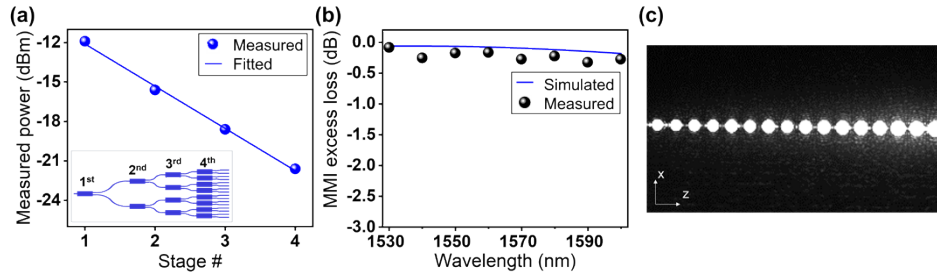


Fig. 7. (a) Measured optical power for output ports at each stage of the MMI coupler. The inset illustrates the schematic of the cascaded 16-channel power splitter. (b) Measured and simulated spectral excess loss of the MMI coupler (c) Near-field guided mode profile of the 16-channel phase modulator array with a spacing of $18\ \mu\text{m}$ at $\lambda = 1550\ \text{nm}$. Intensity is decreased owing to the ohmic loss caused by the metal passing over the waveguide core.

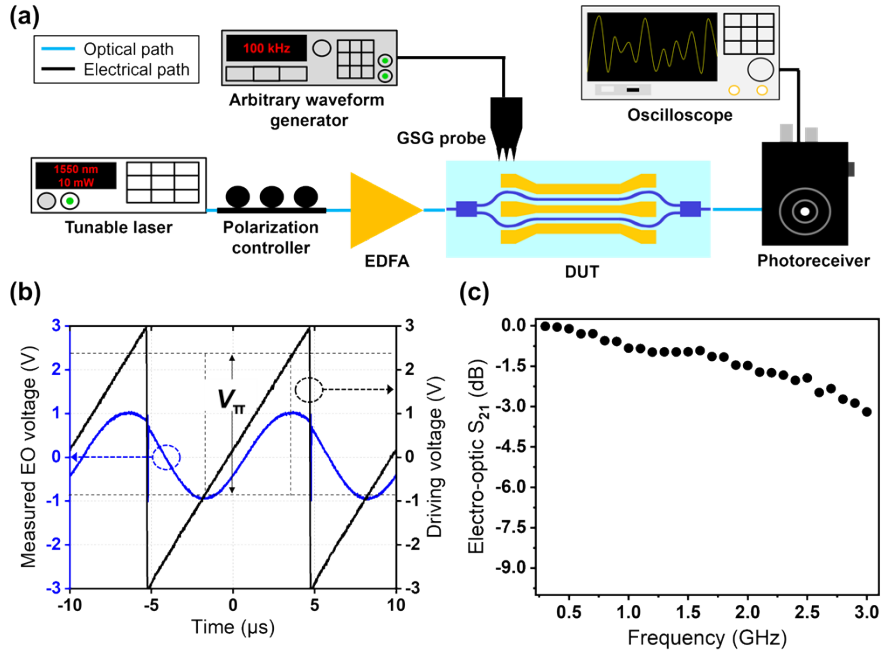


Fig. 8. (a) Experimental setup for characterizing the TFLN–SiN phase modulator. (b) Measured amplitude of the EO signal in the phase modulator as a function of the applied saw voltage. (c) Measured EO response in terms of the frequency for an interferometric modulator with an interaction electrode length of 9.4 mm.

an oscilloscope. The measured EO signal from the photoreceiver is plotted in Fig. 8(b). The measured V_π was 3.025 V for $L_{\text{elec}} = 0.94\ \text{cm}$, leading to a half-wave voltage length product of 2.84 V·cm. For a one-armed phase modulator, the corresponding half-wave voltage length product was 5.68 V·cm, which could be ameliorated by narrowing the electrode gap. Figure 8(c) shows the measured EO response (S_{21}) of a Mach–Zehnder interferometric modulator for $L_{\text{elec}} = 0.94\ \text{cm}$, exhibiting a 3-dB bandwidth of $\sim 2.9\ \text{GHz}$. Considering the RC-limited modulation bandwidth was calculated up to 3.2 GHz for the case of 0.94 cm, the bandwidth of the practical phase modulator with $L_{\text{elec}} = 1.56\ \text{cm}$ was accordingly estimated to be $\sim 1.2\ \text{GHz}$. As shown in Fig. 9(a), the steering performance of the EO modulator array was observed by applying a voltage to the

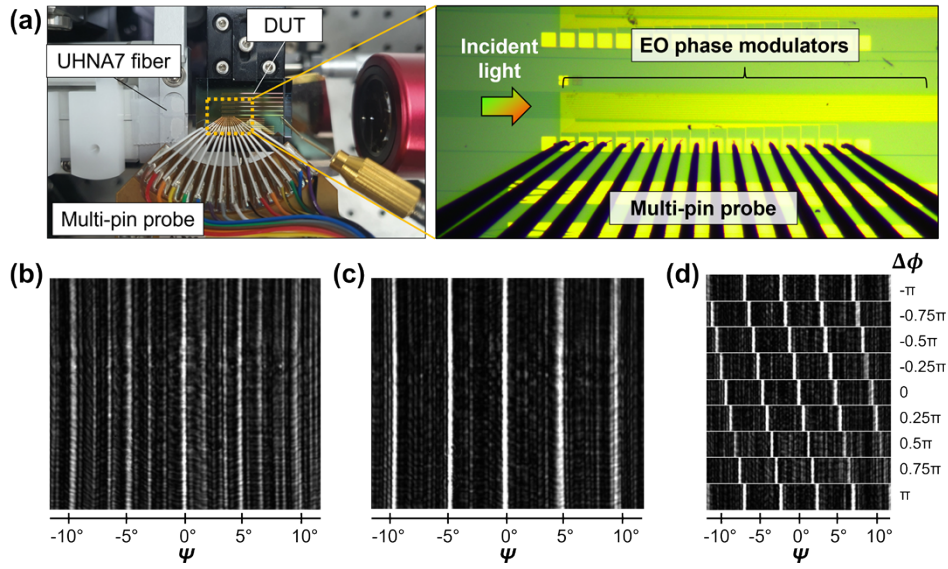


Fig. 9. (a) Experimental setup for the far-field measurement of the TFLN-SiN EO phase modulator array. Phase of each channel is modulated by applying voltage via multi-pin probe. Far-field patterns (b) before and (c) after phase error calibration. (d) Horizontal far-field distribution in response to $\Delta\phi$ between channels.

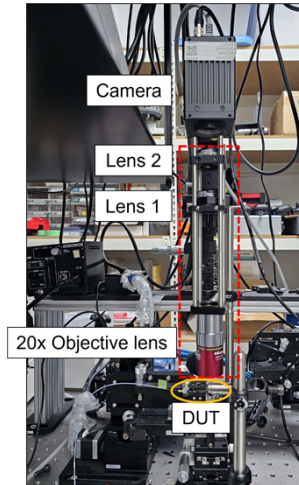


Fig. 10. 4- f lens system for near- and far-field imaging.

phase modulators using a multipin probe. Figure 9(b) shows the far-field patterns radiating from the EOPA facet, captured by the SWIR camera after passing through the objective lens. The far-field patterns appeared to be scattered horizontally due to randomly initiated phases. Resorting to a rapid beamforming algorithm based on the rotating element vector (REV) method was capitalized on to obtain a well-defined beam by compensating for initially arbitrary phase errors [33]. Phase calibration was categorically confirmed to establish highly defined line beams that included the main and strong side lobes among the scattered far-field patterns, as shown in Fig. 9(c). The side lobes were determined in accordance with the theoretical relationship

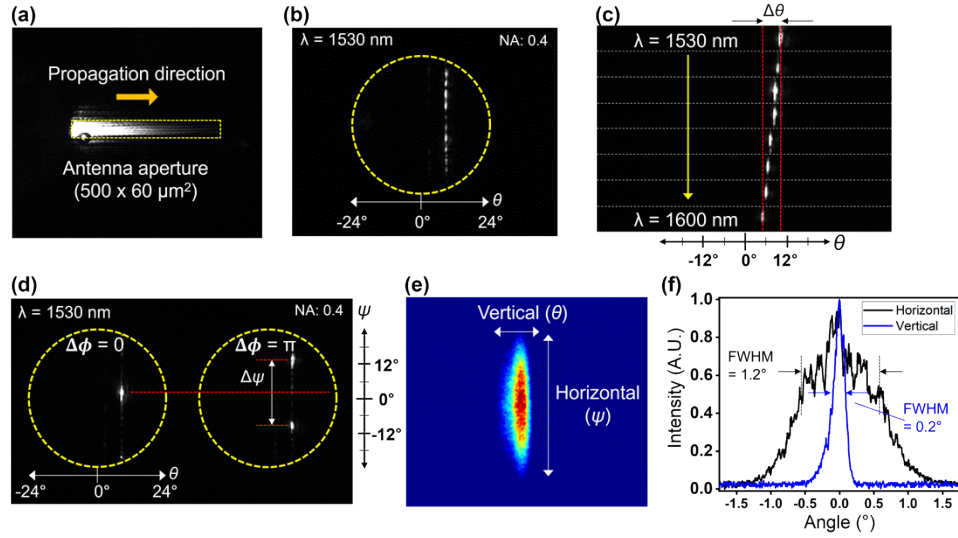


Fig. 11. (a) Captured near-field profile or pattern of the SiN grating emitter with an aperture of $500 \times 60 \mu\text{m}^2$. (b) Far-field pattern without phase calibration at a wavelength of 1530 nm. Far-field distribution in response to (c) wavelengths varying from 1530 to 1600 nm at 10 nm intervals along the vertical direction and (d) phase tuning along the horizontal direction at $\lambda = 1530$ nm. (e) Captured main lobe beam spot emitted from the grating antenna. (f) Measured cross-sectional profiles of the main lobe to determine the beamwidth.

$\psi = m\sin^{-1}(\lambda\Delta\phi/2\pi d)$ where m is an integer, $\Delta\phi$ is the phase difference between consecutive channels, and d is the channel spacing. The far-field patterns were monitored when $\Delta\phi$ between adjacent waveguides linearly varied from $-\pi$ to π for steering the beam horizontally, as shown in Fig. 9(d). The measured scanning range was from -2.5° to 2.5° , yielding a full steering range of 5° . However, the vertical beam divergence was too large to capture owing to the small aperture along the vertical direction. The defined line beams, which were not consistently maintained, deteriorated over time owing to the direct-current (DC) drift phenomenon—a prevalent issue in lithium niobate modulators. The DC drift can be mitigated by thermal treatment after removing the modulator claddings [34]. Even in the presence of DC drift, normal functionality was sustainable via faster beam steering than the DC drift response. To examine the electrical power dissipation, an LCR meter (HIOKI, IM3536) was adopted to measure the capacitance of each EO phase modulator as 0.98 pF. Thus, the power consumption was accordingly $3.2 \text{ pJ}/\pi$ per phase modulator, under the measured V_π of 3.64 V for $L_{\text{elec}} = 1.56 \text{ cm}$. The SiN grating antenna was appended to the EO modulator array to provide a substantially enhanced FOV and beam quality, in conjunction with the ability of wavelength-tuned beam steering. As a result of evaluating the optical loss characteristics of its constituting components, the total insertion loss of the proposed EOPA was approximately 17.3 dB at 1550-nm wavelength, which is accounted for by the fiber-to-chip coupling loss (4.3 dB), insertion loss of the MMI power splitter (0.7 dB), propagation loss (5.1 dB for 2.5-cm length), TFLN–SiN chip-to-SiN chip coupling loss (1.5 dB), and emission loss associated with the SiN grating antenna (5.7 dB). Figure 10 presents a 4- f lens system in combination with a vertically standing near- and far-field measurement setup for characterizing the proposed EOPA beamforming device. We tapped into an objective lens with the numerical aperture of 0.4 corresponding to an acceptable FOV of 48° to obtain high-performance far-field images. The focal lengths of lenses 1 and 2 were 100 and 50 mm, respectively. The measurement setup alternated between near- and far-field images by installing or removing lens 1. A near-field image of the SiN grating emitter was captured using the SWIR camera, as shown in Fig. 11(a).

Table 1. Performance comparison of the reported high-speed OPAs

Platform	Modulation mechanism	FOV	Beamwidth	Power consumption	Bandwidth	Ref.
Silicon	Charge-injection	$8.9^\circ \times 2.2^\circ$	$0.92^\circ \times 0.32^\circ$	17 mW/ π	324 MHz	[37]
EO polymer	Pockels effect	19.1°	3.9°	47.5 μ W/ π	2 MHz	[13]
InP	Carrier injection	8.88°	0.11°	48 mW/ 2π	<16 ns	[12]
TFLN	Pockels effect	$24^\circ \times 8^\circ$	$2^\circ \times 0.6^\circ$	13.5 pJ/ π^a	4.2 GHz ^a	[23]
TFLN	Pockels effect	$50^\circ \times 8.6^\circ$	$0.73^\circ \times 2.8^\circ$	330 pJ/ π	2.5 GHz	[24]
TFLN–SiN	Pockels effect	$22^\circ \times 5^\circ$	$1.2^\circ \times 0.2^\circ$	3.2 pJ/ π	1.2 GHz	This work

^aSimulation result.

The radiation was stronger in front of the antenna compared to its rear end; hence, the effective aperture was less than the designed size of $500 \times 60 \mu\text{m}^2$. From the perspective of enlarging the effective aperture, a uniform emission profile could be secured by apodizing the perturbation ratio [35,36]. Figure 11(b) shows the captured far-field intensity pattern of the 1530 nm-wavelength light. The absence of phase calibration was seen to engender several side lobes and limited beam contrast along the horizontal direction. After phase error correction counting on the REV method, a highly efficient beamforming could be accomplished. Figure 11(c) shows that the wavelength was tuned from 1530 to 1600 nm to adjust the emission angle along the vertical direction. This gave rise to a steering range $\Delta\theta$ of 10.3° to 5.3° , which coincided with the expected tuning efficiency of $0.07^\circ/\text{nm}$. Subsequently, far-field distribution was explored in the horizontal direction as a function of $\Delta\phi$ within $-\pi$ and π , as shown in Fig. 11(d). The steering range $\Delta\psi$ from -11° to 11° was realized, with the side lobes manifested at the angle of 22° away from the main lobe. The beamwidth was not consistently stable due to the relaxation time (several tens of milliseconds) of the modulator, which is yet faster than the capturing time. To mitigate the instability of the beam spot, the adoption of annealing process after removing the cladding atop the electrode could be taken into account [34]. It is remarked that deploying a driver exhibiting a fast bandwidth of tens of MHz should be able to unequivocally suppress the adverse influence of the photorefractive effect [34,37]. The beamwidth has been monitored with the aid of a separate SiN grating antenna, which was equivalent to the structure deployed in the proposed component. As shown in Fig. 11(e), the main beam spot of the antenna was observed with a beam profiler (CINOGY, CMOS-1202). According to the observed cross-sectional beam profiles displayed in Fig. 11(f), the full width at half maximum beamwidths were 1.2° and 0.2° along the horizontal and vertical direction, respectively. The FOV and beamwidths can be readily adjusted by optimizing the design of the grating and phase modulator array and by taking advantage of an optical lens or metalens [5,10]. For a TFLN–SiN photonic platform-based EO modulator array, the optical mode confinement may be weakened to potentially degrade the beam quality for beamforming. To mitigate this issue, an SiN photonic platform can be recommended as an antenna leading to well-formed beams, thereby improving the FOV. Table 1 summarizes the key performance metrics of the proposed OPA in comparison with previously reported devices. Our OPA could be legitimately judged to provide a prominent performance in terms of the power consumption and modulation bandwidth. The proposed EOPA, featuring high-speed beam steering with low power consumption, is anticipated to usher in diverse applications, where a hybrid integration of active and passive components is facilitated without directly etching TFLN [38].

4. Conclusion

This study developed a hybrid TFLN–SiN EOPA with an SiN grating antenna to facilitate high-speed and low-power 2D beam steering. The 16-channel EO modulator array exhibited an estimated modulation speed of 1.2 GHz, power consumption of $3.2 \text{ pJ}/\pi$ per modulator with a V_π value of 3.64 V, and horizontal steering range of 5° . To improve the FOV, beam quality, and vertical beam steering capability, an SiN platform was used instead of a TFLN–SiN platform with weak mode confinement as the proposed device antenna. The SiN grating antenna was butt coupled to the EO modulator array to realize the EOPA. An experimental demonstration of our approach was established in the wavelength range from 1530 to 1600 nm. The EOPA achieved a steering range of $22^\circ \times 5^\circ$ with the full width at half maximum beamwidth of $1.2^\circ \times 0.2^\circ$ along the phase and wavelength tuning directions. The steering performance of the proposed approach, which involved GHz modulation and low power consumption, can be further improved by engineering a grating antenna and array. It is anticipated that our device will be a pivotal key element in many applications such as high-speed imaging, LiDAR, and optical wireless communication.

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Data availability. Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

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